

Course Syllabus

Becoming a Data Scientist

Introduction

What is Data Science? What does it mean to be a Data Scientist? This course aims to help students achieve a deeper understanding of the ins and outs of this field with real world case scenarios and examples. Students will be able to facilitate conversations with instructors regarding the specialisation courses typically offered in the third semester to get an insight into what that field entails. Students will also encounter simulations and case studies that will present them with ethical dilemmas and they will learn to navigate ethical considerations of the work they do and the information they provide. Lastly, this course will enable students to competently communicate the findings and observations of their data analyses and modelling with both technical as well as non-technical audiences.

Meeting Dates & Times

Each week we will meet for a 1.5 hour session. Our time together in class will involve a mix of lecture content, guest speakers, in-class skill and team building activities, and time to work on your team-based semester-long projects.

Instructors & Office Hours

Course email: si568-instructors@umich.edu (<mailto:si568-instructors@umich.edu>)

LINK TO OFFICE HOURS [📄 \(https://go.oncehub.com/SI568OH\)](https://go.oncehub.com/SI568OH)

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Adjunct Lecturer, School of Information

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Graduate Student Instructor

Nancy Xu - yinuoxu@umich.edu

Graduate Student Instructor

High Level Learning Goals

- Students are introduced to the career of Data Science.
- Students will be able to write technical reports.
- Students will be competent in communicating results to technical and non-technical stakeholders.
- Students will be able to navigate various ethical considerations and make ethical choices.
- Students will be aware of the various subfields of Data Science.

Grade Distribution & Grade Conversion

In Class Labs **20%**, Homework Assignments **20%**, Projects **60%**

Letter Grade	Percentage Grade		Letter Grade	Percentage Grade
A+	99-100%		C+	77-79.9%
A	93-98.9%		C	73-76.9%
A-	90-92.9%		C-	70-72.9%
B+	87-89.9%		D+	67-69.9%
B	83-86.9%		D	63-67.9%
B-	80-82.9%		D-	60-62.9%

Weekly Assignments

There are various assignment types that may be assigned each week. The distribution of points may differ slightly based on that week's topic. Please make sure to check each module for each week's homework. Individual Programming assignments should take 3 hours, if you are not going to complete the work in 3 hours, stop at 2.5 hours and write a reflection.

Additionally, lab assignments maybe assigned during class. Labs are meant to be completed during class. However, submission will be accepted till **Fridays at 10am**.

Academic Integrity

Violation of Academic Integrity - whether plagiarism or otherwise - will be penalised. Students are expected to uphold themselves to the highest standards of professional conduct. The first incident of violation of academic integrity will result in an F grade for all involved students on that particular assignment(s). Any subsequent incident will result in an automatic failure of the course.

Collaboration

UMSI strongly encourages collaboration while working on some assignments, such as homework problems and interpreting reading assignments as a general practice. Active learning is effective. Collaboration with other students in the course will be especially valuable in summarizing the reading materials and picking out the key concepts. You must, however, write your homework submission on your own, in your own words, before turning it in. If you worked with someone on the homework before writing it, you must list any and all collaborators on your written submission. Each course and each instructor may place restrictions on collaboration for any or all assignments. Read the instructions carefully and request clarification about collaboration when in doubt. Collaboration is almost always forbidden for take-home and in-class exams.

Plagiarism

All written submissions must be your own, original work. Original work for narrative questions is not merely paraphrasing someone else's completed answer: you must not share written answers with each other at all. At most, you should be working from notes you took while participating in a study session. Largely duplicate copies of the same assignment will receive an equal division of the total point score from the one piece of work. You may incorporate selected excerpts, statements or phrases from publications by other authors, but they must be clearly marked as quotations and must be attributed. If you build on the ideas of prior authors, you must cite their work. You may obtain copy editing assistance and discuss your ideas with others, but all substantive writing and ideas must be your own or be explicitly attributed to another. See the (Doctoral, MSI, BSI) student handbooks available on the UMSI intranet for the definition of plagiarism, resources to help you avoid it, and the consequences of intentional or unintentional plagiarism.

COVID-19 statement

For the safety of all students, faculty, and staff on campus, it is important for each of us to be mindful of safety measures that have been put in place for our protection. By returning to campus, you have acknowledged your responsibility for protecting our community's collective health. Your participation in this course on an in-person basis is conditional upon your adherence to all safety measures mandated by the State of Michigan and the University, including maintaining a physical distance of six feet from

others. The instructor will be wearing a face mask, and while it is not mandatory for students to do so, it is encouraged. The instructor also requests that face masks be worn if students are going to approach the instructor within 6 ft distance or while attending in-person office hours. Your ability to participate in this course in person and your grade may be impacted by failure to comply with campus safety measures. If you are unable or unwilling to adhere to these safety measures while in a face-to-face class setting, you will be required to participate on a remote basis (if available) or to dis-enrol from the class.

Disability Statement

The University of Michigan recognizes disability as an integral part of diversity and is committed to creating an inclusive and equitable educational environment for students with disabilities. Students who are experiencing a disability-related barrier should contact [Services for Students with Disabilities](https://ssd.umich.edu/) (<https://ssd.umich.edu/>); 734-763-3000 or ssdoffice@umich.edu). For students who are connected with SSD, accommodation requests can be made in Accommodate. Please contact your SSD Coordinator or visit SSD's Current Student webpage if you have any questions or concerns. SSD considers aspects of the course design, course learning objects and the individual academic and course barriers experienced by the student. Further conversation with SSD, instructors, and the student may be warranted to ensure an accessible course experience. The instructional team will treat any information that you provide in as confidential a manner as possible.

Audio and Video Recording

Class Recordings

We will be doing audio and video recording of all sessions to enable those who cannot attend class in person on a given day to access the content. Recordings of all sessions will be available on Canvas only to students registered for this class. As part of your participation in this course, you may be recorded. If you do not wish to be recorded, please contact the professor during the first week of class to discuss alternative arrangements. The camera only picks up the front of the room (instructor and slides), but this may require you to sit in a particular place in the room outside the camera's view. Our classroom does have a ceiling mike that picks up student voices. Students may not copy and share the lecture videos with those not in the class, or upload them to any other online environment (this is a violation of the Federal Education Rights and Privacy Act (FERPA)).

Personal recordings are prohibited.

If you are a pregnant, parenting student, or primarily responsible for providing care for a loved one or family member, and you are in need of any accommodations, please let the instructor know at your earliest convenience. You may also reach out to mcasep.org and CEW+ for resources and community support.

Student Mental Health and Wellbeing

Students may experience stressors that can impact both their academic experience and their personal well-being. These may include academic pressure and challenges associated with relationships, mental health, alcohol or other drugs, identities, finances, etc.

If you are experiencing concerns, seeking help is a courageous thing to do for yourself and those who care about you. If the source of your stressors is academic, please contact me so that we can find solutions together. For personal concerns, U-M offers the following resources:

- **Counselling and Psychological Services (CAPS)** (<https://caps.umich.edu/>) - confidential; 734-764-8312; for after-hours urgent support, call and press 0; counselling, workshops, groups and more. Ashley Ewearitt, a CAPS counsellor, is embedded in UMSI; information about how to schedule an appointment with her can be found [here](https://docs.google.com/document/d/1EWAfqqK_S1VMV9OyToz5gsUUP9oqXXNSBjcKviY_XVE/edit) (https://docs.google.com/document/d/1EWAfqqK_S1VMV9OyToz5gsUUP9oqXXNSBjcKviY_XVE/edit).
- **Dean of Students Office** (<https://deanofstudents.umich.edu/>) - 734-764-7420; provides support services to students and manages critical incidents impacting students and the campus community.
- **Ginsberg Center for Community Service Learning** (<https://ginsberg.umich.edu/>) - 734-763-3548; opportunities to engage as learners and leaders to create a better community and world
- **Maize and Blue Cupboard (MBC)** (<https://mbc.studentlife.umich.edu/>) - 734-936-2794; Food pantry with groceries, kitchen and cooking supplies, personal and household items, and support
- **Multi-ethnic Student Affairs (MESA)** (<https://mesa.umich.edu/>) - 734-763-9044; diversity and social justice through the lens of race and ethnicity
- **Office of Student Conflict Resolution** (<https://oscr.umich.edu/>) - 734-936-6308; offers multiple pathways for resolving conflict.
- **Office of the Ombuds** (<https://ombuds.umich.edu/>) - 734-763-3545; students can raise questions and concerns about the functioning of the university.
- **Services for Students with Disabilities (SSD)** (<https://ssd.umich.edu/>) - 734-763-3000; accommodations and access to students with disabilities
- **SilverCloud** (<https://caps.umich.edu/silvercloud>) - Students may also use an online, self-guided, interactive mental health resource that provides cognitive behavioural interventions.
- **Sexual Assault Prevention and Awareness Center (SAPAC)** (<https://sapac.umich.edu/>) - confidential; 734-764-7771 or 24-hour crisis line 734-936-3333; addresses sexual assault, intimate partner violence, sexual harassment, and stalking.
- **Spectrum Center** (<https://spectrumcenter.umich.edu/>) - 734-763-4186; support services for LGBTQ+ students
- **Trotter Multicultural Center** (<https://trotter.umich.edu/>) - 734-763-3670; intercultural engagement and inclusive leadership education initiatives
- **University Health Service (UHS)** (<https://uhs.umich.edu/>) - 734-764-8320; clinical services include nurse advice by phone, day or night

- **Well-being for U-M Students website** (<https://wellbeing.studentlife.umich.edu/>) - searchable list of many more campus resources
- **Wolverine Wellness** (<https://uhs.umich.edu/wolverine-wellness>) - 734-763-1320; provides Wellness Coaching, Collegiate Recovery Program, and much more